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Unit – 4 : Session – 10 : SLO – 1 | Odd Man Out — Pipelining Performance

■ I. Odd Man Out

1. Group CPI, IPC, Speedup, Bandwidth → Bandwidth — a data-transfer rate; the others are instruction-level performance metrics.
2. Group Throughput, Latency, Clock rate, Hard disk RPM → Hard disk RPM — a storage-device spec, not a CPU pipeline metric.
3. Group Ideal speedup, Observed speedup, Pipeline depth, Cache size → Cache size — a memory-hierarchy parameter, not a pipeline speedup/depth measure.
4. Group Data forwarding, Branch prediction, Hazard elimination, Motherboard → Motherboard — a physical board; the others are pipeline hazard techniques.
5. Group Structural, Control, Data hazard, Power outage → Power outage — an external failure, not a pipeline hazard.
6. Group Superpipelining, Superscalar, VLIW, Solid-state drive → Solid-state drive — storage; the others are ILP architectures.
7. Group Bubble, Flush, Stall, Acceleration → Pipeline acceleration — the others are pipeline disruption terms.
8. Group ILP, Thread-level parallelism, Sequential processing → Sequential processing — no parallelism.
9. Group 1 CPI, 0.5 CPI, 2 CPI, 5 GHz → 5 GHz — a clock frequency; the others are cycles-per-instruction values.
10. Group RISC pipeline, CISC pipeline, Printer driver → Printer driver — peripheral software, not a processor pipeline.